SYNCHRONY

Real-time hybrid fraud decision support

Digital lending · explainable risk · human oversight

ROLL NUMBER se23ucse120

Prototype using synthetic data. HIGH_RISK means investigate—not deny credit.

From isolated applications to connected fraud signals

A real-time investigator workflow, not another opaque binary classifier

01 · THE GAP

Static rules miss new patterns. A supervised model alone struggles with novel behavior, bursts, and identities connected by shared devices, IPs, or bank accounts.

02 · THE RISK

False positives delay access to credit; false negatives create loss. The system must expose evidence, preserve privacy, and keep a human in control.

03 · THE OUTCOME

One 0–100 risk score, three operational bands, four transparent components, curated reasons, investigator feedback, and a live dashboard.

Design question

How can we detect known fraud, novel anomalies, suspicious velocity, and coordinated rings—without turning a prototype into an autonomous lending gate?

A modular, API-first vertical slice

The simulator and dashboard use the same production-shaped assessment path

React → **FastAPI** → **Features** → **Hybrid AI** → **Risk policy** → **SQL**

React	FastAPI	Features	Hybrid AI	Risk policy	SQL
monitor · investigate	validate · authorize	point-in-time · v1	4 component scores	score · explain	persist · analyze

LOCAL

SQLite + locked in-memory prior state. One-command startup keeps judging and demos reliable.

DEPLOYMENT

PostgreSQL 16 + atomic Redis state in non-root Docker containers with readiness checks.

SECURITY

Reader/admin API keys, HMAC pseudonyms, request limits, restricted CORS, and environment-only secrets.

Optional AI stack: Temporal tabular scoring—not semantic search or generation. Deterministic models are safer and measurable here.

Honest data boundaries before impressive metrics

PaySim supplies labels; lending context is deterministic synthetic enrichment with field-level provenance

- Canonical strict event schema; identifiers and labels never enter the 26-feature model contract.
- Chronological train / validation / test split before fitting preprocessing or selecting thresholds.
- Offline replay uses the same state semantics as online inference: snapshot prior events, score current event, then record it.
- A manifest records source, hash, seed, time boundaries, schema, row counts, and entity-overlap checks.
- The bundled 600-row development fixture proves the pipeline—it is not representative lending evidence.

SOURCE

PaySim synthetic mobile-money transactions and their fraud label.

ENRICHMENT

Loan, income, device, login, geography, and application-history features generated deterministically.

PARITY INVARIANT

No current/future event can leak into velocity or shared-entity features.

Four specialists, one configurable decision

Each component emits a normalized 0–1 score and safe signal codes

38% · SUPERVISED

XGBoost challenger selected against a class-weighted logistic baseline using validation PR-AUC and operating constraints.

12% · ANOMALY

Isolation Forest detects unusual combinations; empirical calibration maps raw scores into a stable 0–1 range.

35% · BEHAVIOR

Velocity, failed login, device transition, amount deviation, and loan-to-income risk signals.

15% · GRAPH

Shared device, IP, and bank-account cardinality expose coordinated identity rings without a heavy graph platform.

0 — **40** — **70**

APPROVE — MANUAL REVIEW — HIGH RISK

Actionable evidence without leaking fraud logic

Raw model contributions are translated through a curated reason catalog

MODEL CONTRIBUTION

SHAP for the tree model (or coefficients for the baseline) identifies stable directional features. Raw values never leave the service boundary.

SIGNAL CATALOG

Model and rules map to controlled codes such as HIGH_LOAN_INCOME_RATIO, APPLICATION_BURST, and SHARED_DEVICE_IDENTITIES.

INVESTIGATOR VIEW

The API returns ranked human-readable reasons, four component scores, decision, timestamp, and versioned audit metadata.

Example high-risk explanation

1. Multiple applicant identities share this device.
2. Application activity increased sharply within a short window.
3. Requested amount is high relative to reported annual income.

TRANSPARENCY BOUNDARY

No raw identifiers, sensitive proxy explanations, user-supplied text, contribution magnitudes, or internal rule thresholds.

Monitor, investigate, and replay—not just call a model

React + TanStack Query + Recharts; server state remains the source of truth

Synchrony / Monitoring

TOTAL	APPROVE	REVIEW	HIGH RISK
11	1	4	6

LIVE APPLICATION FEED

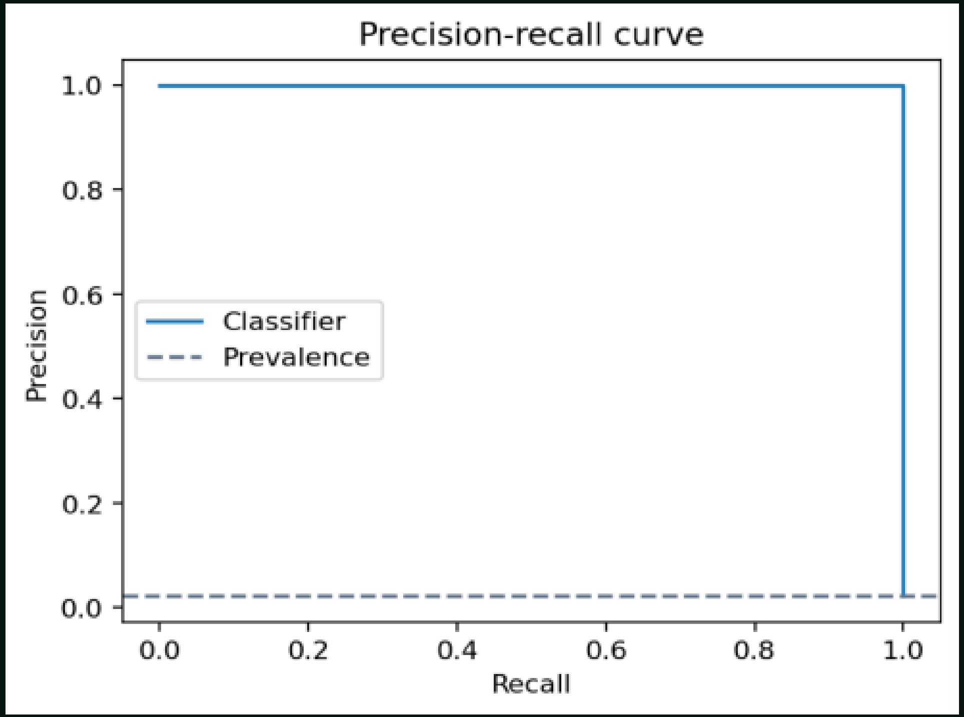
APP-NORMAL-01	18	APPROVE
APP-BURST-02	57	MANUAL REVIEW
APP-RING-03	84	HIGH RISK

DEMO PROOF

- Seeded scenarios span web, mobile, partner API, and agent channels.
- Manual entries and seeded events call the same real /predict path.
- One-second polling shows persisted results.
- Investigations expose an action and verified-outcome controls.
- Reset and replay are deterministic and tested.

Promising evidence—with the caveat in the headline

Chronological test partition of the deliberately small 600-row development fixture



RECALL

1.00
Classifier at frozen threshold

FPR

0.0909
Classifier only

HYBRID RECALL

1.00
Flagged band

HYBRID FPR

0.00
Designed fixture

INTERPRETATION

These numbers verify code, leakage controls, and demo separability. They do not estimate real lending performance.

Human oversight is part of the architecture

Safety controls are product behavior—not a disclaimer added at the end

- No race, religion, gender, or other protected traits are model inputs; geography is excluded from public reasons and segment claims.
- A broad MANUAL_REVIEW band avoids turning uncertainty into automatic rejection.
- Every decision includes model, risk-config, and feature-schema versions plus curated reasons.
- Verified outcomes measure false positives and missed fraud; enough evidence can recommend an offline retraining review, never a silent live update.
- Segment reports include support counts and suppress interpretation where samples are sparse.
- Known risks remain: domain shift, concept drift, label bias, false-positive harm, and unmeasured intersectional disparities.

SYSTEM OUTPUT

Investigation priority and evidence

HUMAN ACTION

Review context, verify identity, document outcome

GOVERNED DECISION

Credit outcome, notice, and appeal outside this prototype

Minimize data, fail closed, separate privileges

Prototype controls are explicit; production upgrades are documented rather than implied

ACCESS

Production requires distinct 32+ character reader and administrator keys. Reader is GET-only; admin may score and control demos.

PRIVACY

HMAC pseudonyms and a derived-field allowlist; no raw user, device, IP, bank, income, amount, or request body in storage/logs.

API HARDENING

Strict schema and bounds, 64 KiB body cap, safe error responses, request IDs, constrained CORS, parameterized ORM access.

SUPPLY CHAIN

Pinned dependency ranges, artifact SHA-256/schema checks before deserialization, and no automatic dataset downloads.

RUNTIME

Non-root containers, no-new-privileges, service health checks, internal-only Redis, environment-backed secrets.

NEXT GATE

Enterprise OIDC, short-lived tokens, TLS termination, immutable audit logs, secret manager, rate limits, SIEM/telemetry.

Small contracts keep the difficult parts testable

Training replay and online requests converge on the same feature and decision boundaries

```
event -> state.snapshot(event, now)
features = FeatureBuilder().transform(event, snapshot)
components = {
  "supervised": classifier.score(features),
  "anomaly": anomaly.score(features),
  "behavioral": behavior.score(features),
  "graph": graph.score(features),
}
assessment = risk_engine.combine(components)
repository.save(event, assessment)
state.record(event, assessment)
```

- Strict LoanApplicationEvent and FraudAssessment schemas define the transport contract.
- Prior-only snapshot ordering prevents self-counting and future leakage.
- Scorers are independently swappable and return normalized, typed outputs.
- Risk weights and thresholds live in validated YAML—not frontend logic.
- Repository and state Protocols support local and deployed adapters.
- Artifact loaders verify digest, size, schema, version, and estimator type.

Evidence gates, not screenshot-driven confidence

The project is reproducible locally and production-shaped in Compose

BACKEND

125 tests including feedback + random stream
81% backend/training coverage
Ruff clean

FRONTEND

Vitest component suite
TypeScript production build
0 npm audit vulnerabilities

END TO END

Real artifacts loaded
11-event replay deterministic
HTTP/API and bundle smoke checks

- README: one-command bootstrap, local run, full-data training, Compose, tests, and troubleshooting.
- Versioned model manifests and generated evaluation, hybrid, plot, and responsible-AI reports.
- Docker Compose: React, FastAPI, PostgreSQL, Redis, health checks, non-root processes.
- GitHub main is committed and pushed; the private repository is ready to share with reviewers or make public.

Verification counts should be refreshed immediately before submission; see docs/VERIFICATION.md.

What moves this from prototype to controlled pilot

Every next step is an evidence gate, not a feature wish list

01 REAL DATA

Representative lending labels, consent and retention review

02 VALIDATE

Calibration, temporal backtests, cost curves, fairness uncertainty

03 GOVERN

OIDC, roles, appeal workflow, model approval and rollback

04 OPERATE

Drift/SLO monitoring, rate limits, audit logs, incident runbook

05 PILOT

Shadow mode, investigator feedback, controlled threshold changes

KEEP THE CORE INVARIANT

Point-in-time data → transparent components → configurable policy → human decision

SYNCHRONY

Detect connected risk.
Explain the evidence.
Keep humans accountable.

React · FastAPI · PostgreSQL · Redis · XGBoost · Isolation Forest · SHAP

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